

Qingguo Qi

Ph.D. Student, ZJU · Efficient LLM Architectures and Long Context · Research Internship (Hangzhou; immediate start; 4-5 days/week; 6+ months)

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Education

Zhejiang University — College of Computer Science and Technology (Ph.D.)

Hangzhou, China

Efficient LLMs and long context; ACL 2026/KDD 2025 first author (CCF-A); expected graduation: 2028

Sep. 2024–Present

UCAS — Hangzhou Institute for Advanced Study (Master's)

Hangzhou, China

Research: dynamic GNNs and large-scale graph learning; Outstanding Graduate

Sep. 2021–Jun. 2024

Tongji University — School of Software Engineering (Bachelor's)

Shanghai, China

Second-Class Academic Scholarship

Sep. 2011–Jun. 2015

Publications and Invention Patents

- **QingGuo Qi** et al. SCOPE: Boosting LLM Efficiency with Scoped Position Encoding. ACL 2026 (CCF-A; first author).
- **QingGuo Qi** et al. Input Snapshots Fusion for Scalable Discrete-Time Dynamic Graph Neural Networks. KDD 2025 (CCF-A; first author).
- Prediction Method, Apparatus, Readable Storage Medium and Electronic Device; CN202310405190.4 (granted; first inventor).
- Time-Series Prediction and Model Training Methods Based on 3D Fully Connected Fusion; CN202311702608.4 (granted; first inventor).

Research Experience

SCOPE: Implicit Position Encoding and Efficient Long-Context Attention

ZJU / Zhejiang Lab

Ph.D. Student · First Author

Sep. 2024–Present

- Proposed multi-scale implicit position encoding via structured sparsity for efficient modeling and length extrapolation.
- Theoretical and empirical analyses show hierarchical sequence processing, with order-awareness horizons growing exponentially with depth without explicit positional signals.
- Pretrained the LLaMA-3-8B architecture from scratch on 50B tokens with TorchTitan; extended context 4K→32K→128K and implemented FlexAttention/Triton training and inference kernels.
- Attention FLOPs: about 1/8 of dense (theoretical, selected settings); 256K end-to-end prefill: about 2× RoPE speed.
- 128K needle-in-a-haystack: 92% mean accuracy (RoPE: 86%), showing long-range retrieval with sparse computation.

SFDyG: Scalable Discrete-Time Dynamic Graph Neural Networks

UCAS / Zhejiang Lab

Master's / Ph.D. Student · First Author

Jul. 2022–Aug. 2025

- Proposed input snapshot fusion into temporal multigraphs to reduce redundant computation and GPU memory overhead.
- Designed Hawkes-based time-decayed message passing for graph structure and temporal dependencies, with mini-batches.
- Evaluated link prediction on 8 public datasets; on Stack Overflow, full-batch MRR@100 improved by about 26.6% over the strongest baseline, while mini-batching cut GPU memory use by about 44% versus full-batch training.

Work and Internship Experience

Zhejiang Lab

Intern / Joint Ph.D. Study

LLM Architecture / AI4Science (Efficient LLMs, Graph Machine Learning)

Jul. 2022–Present

- OGB-LSC NeurIPS 2022: R-GAT/PCA/SGC for large heterogeneous-graph paper classification; team ranked 5th worldwide.
- Proposed a 3D fully connected mixing network for dynamic-graph node attribute prediction, fusing features, space and time to improve prediction quality and efficiency; patent granted.

Dianping, Xiaohongshu and Other Internet Companies

Shanghai, China

Data Engineering

2015–2021

- Processed app business data with Hive/Hadoop and produced visual reports.

Skills

Skills Python, C++; PyTorch, Transformers, TorchTitan; FlexAttention, Triton; PyG, Git, Linux

English CET-6: 582; reading and writing research papers in English

Honors and Awards

2022.11 OGB-LSC NeurIPS 2022 Large-Scale Graph Learning Challenge (Paper Classification)

5th Worldwide

2022.12 Graduate Academic Scholarship, Hangzhou Institute for Advanced Study

First Class

2024.05 Outstanding Graduate, Hangzhou Institute for Advanced Study

UCAS

2025.10 Outstanding Graduate Student, College of Computer Science and Technology

ZJU